

# Bayesian GAMs: R Comparison (brms + mgcv)

GAM.jl Contributors

## Overview

This vignette compares Bayesian GAM posterior distributions from **GAM.jl** (Turing.jl MCMC) with two R approaches:

1. **brms** — Full MCMC via Stan's NUTS sampler. `brms` uses `mgcv::smoothCon()` + `smooth2random()` internally, making it the most direct R analog of GAM.jl's Turing backend. Both use the same `smooth2random` decomposition with half-normal/exponential priors on smoothing SDs.
2. **mgcv Bayesian posterior approximation** — `mgcv::gam()` computes a Bayesian posterior covariance matrix  $V_p$  alongside frequentist estimates. We sample from this multivariate normal posterior.

## Setup

```
library(mgcv)
library(brms)
library(MASS)

# Julia posterior samples (from Turing.jl MCMC)
julia_gauss <- read.csv("../posteriors_gaussian_julia.csv")
julia_poisson <- read.csv("../posteriors_poisson_julia.csv")

# brms posterior samples (from Stan MCMC)
brms_gauss <- read.csv("../posteriors_gaussian_brms.csv")
brms_poisson <- read.csv("../posteriors_poisson_brms.csv")

cat(sprintf("Posterior samples: Julia=%d, brms=%d\n",
            nrow(julia_gauss), nrow(brms_gauss)))
```

Posterior samples: Julia=4000, brms=2000

## Example 1: Gaussian GAM

### Frequentist reference (mgcv)

```
dat <- read.csv("../data_bayes_gaussian.csv")
m <- gam(y ~ s(x, k = 10), data = dat, method = "REML")
summary(m)
```

Family: gaussian

Link function: identity

Formula:

$y \sim s(x, k = 10)$

Parametric coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.11285 | 0.02216    | -5.093  | 8.38e-07 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

|      | edf   | Ref.df | F     | p-value    |
|------|-------|--------|-------|------------|
| s(x) | 6.847 | 7.937  | 124.6 | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.832 Deviance explained = 83.8%

-REML = 65.726 Scale est. = 0.098196 n = 200

### brms fit

The brms model uses the same priors as GAM.jl: `Exponential(1)` on smooth SDs, `Normal(0, 10)` on fixed effects, and `truncated Normal(0, 2.5)` on :

```
m_brms <- brm(
  y ~ s(x, k = 10),
  data = dat,
  family = gaussian(),
  prior = c(
```

```

    prior(normal(0, 10), class = "b"),
    prior(exponential(1), class = "sds"),
    prior(normal(0, 2.5), class = "sigma", lb = 0)
  ),
  chains = 2, iter = 2000, warmup = 1000, seed = 2024,
  backend = "cmdstanr"
)

```

## mgcv Bayesian posterior approximation

`mgcv::gam()` computes the Bayesian posterior covariance  $\mathbf{V}_p$  (accounting for smoothing parameter uncertainty). We sample from the multivariate normal posterior  $\beta \sim N(\hat{\beta}, \mathbf{V}_p)$  and use a scaled inverse- $\chi^2$  posterior for  $\sigma^2$ :

```

set.seed(2024)
n_samples <- 4000

# Coefficient posterior
beta_post <- mvrnorm(n_samples, coef(m), vcov(m))

# Residual SD posterior
n <- nrow(dat)
p <- sum(m$edf)
df_resid <- n - p
sigma2_post <- df_resid * m$scale / rchisq(n_samples, df = df_resid)
sigma_post <- sqrt(sigma2_post)

```

## Three-way comparison: GAM.jl vs brms vs mgcv

```
cat("Parameter          | GAM.jl (Turing)    | brms (Stan)        | mgcv (approx)      | Freq\n")
```

```
Parameter          | GAM.jl (Turing)    | brms (Stan)        | mgcv (approx)      | Freq
```

```
cat("-----|-----|-----|-----\n")
```

```
-----|-----|-----|-----
```

```
|-----
```

```
cat(sprintf("sigma mean      | %.4f          | %.4f          | %.4f          | %.4f\n",
            mean(julia_gauss$sigma_obs), mean(brms_gauss$sigma_obs),
            mean(sigma_post), sqrt(m$scale)))
```

```
sigma mean      | 0.3148          | 0.3151          | 0.3145          | 0.3134
```

```
cat(sprintf("sigma sd        | %.4f          | %.4f          | %.4f          | \n",
            sd(julia_gauss$sigma_obs), sd(brms_gauss$sigma_obs), sd(sigma_post)))
```

```
sigma sd        | 0.0161          | 0.0161          | 0.0160          |
```

```
cat(sprintf("Intercept mean   | %.4f          | %.4f          | %.4f          | %.4f\n",
            mean(julia_gauss$beta_intercept), mean(brms_gauss$beta_intercept),
            mean(beta_post[, 1]), coef(m)[1]))
```

```
Intercept mean   | -0.1128          | -0.1131          | -0.1123          | -
0.1129
```

```
cat(sprintf("Intercept sd      | %.4f          | %.4f          | %.4f          | \n",
            sd(julia_gauss$beta_intercept), sd(brms_gauss$beta_intercept),
            sd(beta_post[, 1])))
```

```
Intercept sd      | 0.0223          | 0.0223          | 0.0221          |
```

## Kolmogorov-Smirnov tests and ECDF correlation

The KS statistic measures the maximum vertical distance between two ECDFs, while the **pairwise ECDF correlation** evaluates both ECDFs on a common grid and computes their Pearson correlation — a value of 1.0 indicates identical distributions:

```
# Helper: pairwise ECDF correlation on a common grid
ecdf_cor <- function(x, y, n_grid = 500) {
  grid <- seq(min(c(x, y)), max(c(x, y)), length.out = n_grid)
  cor(ecdf(x)(grid), ecdf(y)(grid))
}

# GAM.jl vs brms (full MCMC vs full MCMC - should be very similar)
ks_sigma_jb <- ks.test(julia_gauss$sigma_obs, brms_gauss$sigma_obs)
```

```
Warning in ks.test.default(julia_gauss$sigma_obs, brms_gauss$sigma_obs):  
p-value will be approximate in the presence of ties
```

```
ks_int_jb <- ks.test(julia_gauss$beta_intercept, brms_gauss$beta_intercept)
```

```
Warning in ks.test.default(julia_gauss$beta_intercept,  
brms_gauss$beta_intercept): p-value will be approximate in the presence of ties
```

```
# GAM.jl vs mgcv (full MCMC vs Gaussian approximation)  
ks_sigma_jm <- ks.test(julia_gauss$sigma_obs, sigma_post)
```

```
Warning in ks.test.default(julia_gauss$sigma_obs, sigma_post): p-value will be  
approximate in the presence of ties
```

```
ks_int_jm <- ks.test(julia_gauss$beta_intercept, beta_post[, 1])
```

```
Warning in ks.test.default(julia_gauss$beta_intercept, beta_post[, 1]): p-  
value  
will be approximate in the presence of ties
```

```
# brms vs mgcv  
ks_sigma_bm <- ks.test(brms_gauss$sigma_obs, sigma_post)
```

```
Warning in ks.test.default(brms_gauss$sigma_obs, sigma_post): p-value will be  
approximate in the presence of ties
```

```
ks_int_bm <- ks.test(brms_gauss$beta_intercept, beta_post[, 1])
```

```
Warning in ks.test.default(brms_gauss$beta_intercept, beta_post[, 1]): p-  
value  
will be approximate in the presence of ties
```

```
cat("Comparison          | Param      | KS D    | KS p    | ECDF cor\n")
```

```
Comparison          | Param      | KS D    | KS p    | ECDF cor
```

```
cat("-----|-----|-----|-----|-----\n")
```

```
-----|-----|-----|-----|-----
```

```
cat(sprintf("GAM.jl vs brms      | sigma      | %.4f | %.4f | %.6f\n",
  ks_sigma_jb$statistic, ks_sigma_jb$p.value,
  ecdf_cor(julia_gauss$sigma_obs, brms_gauss$sigma_obs)))
```

```
GAM.jl vs brms      | sigma      | 0.0255 | 0.3512 | 0.999880
```

```
cat(sprintf("GAM.jl vs brms      | intercept | %.4f | %.4f | %.6f\n",
  ks_int_jb$statistic, ks_int_jb$p.value,
  ecdf_cor(julia_gauss$beta_intercept, brms_gauss$beta_intercept)))
```

```
GAM.jl vs brms      | intercept | 0.0205 | 0.6296 | 0.999837
```

```
cat(sprintf("GAM.jl vs mgcv (approx)| sigma      | %.4f | %.4f | %.6f\n",
  ks_sigma_jm$statistic, ks_sigma_jm$p.value,
  ecdf_cor(julia_gauss$sigma_obs, sigma_post)))
```

```
GAM.jl vs mgcv (approx)| sigma      | 0.0228 | 0.2518 | 0.999916
```

```
cat(sprintf("GAM.jl vs mgcv (approx)| intercept | %.4f | %.4f | %.6f\n",
  ks_int_jm$statistic, ks_int_jm$p.value,
  ecdf_cor(julia_gauss$beta_intercept, beta_post[, 1])))
```

```
GAM.jl vs mgcv (approx)| intercept | 0.0325 | 0.0293 | 0.999744
```

```
cat(sprintf("brms vs mgcv (approx) | sigma      | %.4f | %.4f | %.6f\n",
  ks_sigma_bm$statistic, ks_sigma_bm$p.value,
  ecdf_cor(brms_gauss$sigma_obs, sigma_post)))
```

```
brms vs mgcv (approx) | sigma      | 0.0215 | 0.5686 | 0.999921
```

```
cat(sprintf("brms vs mgcv (approx) | intercept | %.4f | %.4f | %.6f\n",
  ks_int_bm$statistic, ks_int_bm$p.value,
  ecdf_cor(brms_gauss$beta_intercept, beta_post[, 1])))
```

brms vs mgcv (approx) | intercept | 0.0220 | 0.5387 | 0.999904

```
cat("\n(ECDF correlation near 1.0 = nearly identical distributions)\n")
```

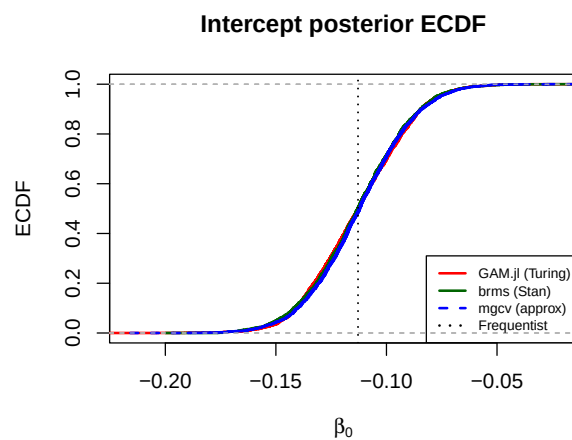
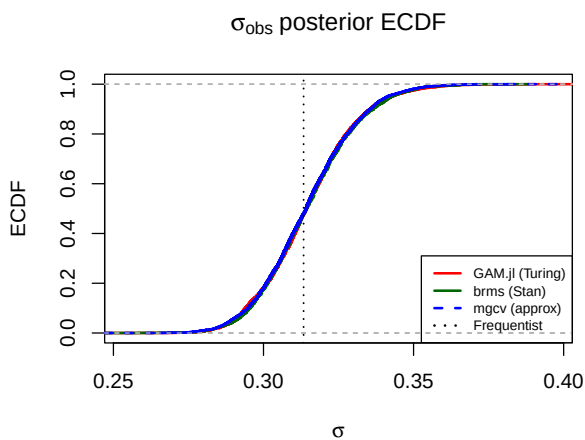
(ECDF correlation near 1.0 = nearly identical distributions)

## ECDF comparison plots

```
par(mfrow = c(1, 2))

# sigma ECDF - three posteriors
plot(ecdf(julia_gauss$sigma_obs), col = "red", main = expression(sigma[obs] ~ "posterior ECDF",
  xlab = expression(sigma), ylab = "ECDF", lwd = 2)
lines(ecdf(brms_gauss$sigma_obs), col = "darkgreen", lwd = 2)
lines(ecdf(sigma_post), col = "blue", lwd = 2, lty = 2)
abline(v = sqrt(m$scale), lty = 3, col = "black", lwd = 1.5)
legend("bottomright", c("GAM.jl (Turing)", "brms (Stan)", "mgcv (approx)", "Frequentist"),
  col = c("red", "darkgreen", "blue", "black"), lty = c(1, 1, 2, 3), lwd = 2, cex = 0.7)

# Intercept ECDF
plot(ecdf(julia_gauss$beta_intercept), col = "red", main = "Intercept posterior ECDF",
  xlab = expression(beta[0]), ylab = "ECDF", lwd = 2)
lines(ecdf(brms_gauss$beta_intercept), col = "darkgreen", lwd = 2)
lines(ecdf(beta_post[, 1]), col = "blue", lwd = 2, lty = 2)
abline(v = coef(m)[1], lty = 3, col = "black", lwd = 1.5)
legend("bottomright", c("GAM.jl (Turing)", "brms (Stan)", "mgcv (approx)", "Frequentist"),
  col = c("red", "darkgreen", "blue", "black"), lty = c(1, 1, 2, 3), lwd = 2, cex = 0.7)
```



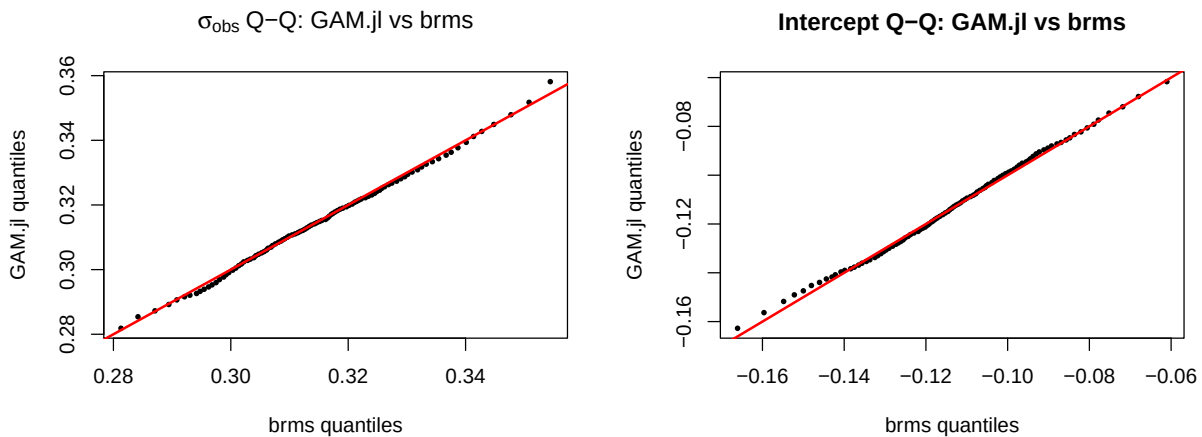
## Q-Q plots: GAM.jl vs brms

The most stringent comparison — both use full MCMC with the same model specification:

```
par(mfrow = c(1, 2))
probs <- seq(0.01, 0.99, by = 0.01)

# sigma Q-Q
q_julia <- quantile(julia_gauss$sigma_obs, probs)
q_brms <- quantile(brms_gauss$sigma_obs, probs)
plot(q_brms, q_julia, pch = 16, cex = 0.6,
     main = expression(sigma[obs] ~ "Q-Q: GAM.jl vs brms"),
     xlab = "brms quantiles", ylab = "GAM.jl quantiles")
abline(0, 1, col = "red", lwd = 2)

# Intercept Q-Q
q_julia_int <- quantile(julia_gauss$beta_intercept, probs)
q_brms_int <- quantile(brms_gauss$beta_intercept, probs)
plot(q_brms_int, q_julia_int, pch = 16, cex = 0.6,
     main = "Intercept Q-Q: GAM.jl vs brms",
     xlab = "brms quantiles", ylab = "GAM.jl quantiles")
abline(0, 1, col = "red", lwd = 2)
```



## Example 2: Poisson GAM

### Frequentist fit



```
dat2 <- read.csv("../data_bayes_poisson.csv")
m2 <- gam(y ~ s(x, k = 10), data = dat2, family = poisson(), method = "REML")
cat(sprintf("Frequentist intercept (log-scale): %.4f (true: 1.0)\n", coef(m2)[1]))
```

Frequentist intercept (log-scale): 0.9094 (true: 1.0)

### mgcv Bayesian posterior

```
beta_post2 <- mvrnorm(n_samples, coef(m2), vcov(m2))
```

### Three-way comparison

```
cat("Parameter          | GAM.jl (Turing)    | brms (Stan)        | mgcv (approx)      | True\n")
```

```
Parameter          | GAM.jl (Turing)    | brms (Stan)        | mgcv (approx)      | True
```

```
cat("-----|-----|-----|-----|-----\n")
```

```
-----|-----|-----|-----|
|-----
```

```
cat(sprintf("Intercept mean    | %.4f          | %.4f          | %.4f          | 1.0\n",
            mean(julia_poisson$beta_intercept), mean(brms_poisson$beta_intercept),
            mean(beta_post2[, 1])))
```

```
Intercept mean    | 0.9083          | 0.9083          | 0.9092          | 1.0
```

```
cat(sprintf("Intercept sd      | %.4f          | %.4f          | %.4f          | \n",
            sd(julia_poisson$beta_intercept), sd(brms_poisson$beta_intercept),
            sd(beta_post2[, 1])))
```

```
Intercept sd      | 0.0529          | 0.0535          | 0.0536          |
```

### KS tests and ECDF

```
ks_jb2 <- ks.test(julia_poisson$beta_intercept, brms_poisson$beta_intercept)
```

Warning in ks.test.default(julia\_poisson\$beta\_intercept, brms\_poisson\$beta\_intercept): p-value will be approximate in the presence of ties

```
ks_jm2 <- ks.test(julia_poisson$beta_intercept, beta_post2[, 1])
```

Warning in ks.test.default(julia\_poisson\$beta\_intercept, beta\_post2[, 1]): p-value will be approximate in the presence of ties

```
ks_bm2 <- ks.test(brms_poisson$beta_intercept, beta_post2[, 1])
```

Warning in ks.test.default(brms\_poisson\$beta\_intercept, beta\_post2[, 1]): p-value will be approximate in the presence of ties

```
cat("Comparison          | KS D    | KS p    | ECDF cor\n")
```

```
Comparison          | KS D    | KS p    | ECDF cor
```

```
cat("-----|-----|-----|-----\n")
```

```
-----|-----|-----|-----
```

```
cat(sprintf("GAM.jl vs brms          | %.4f  | %.4f  | %.6f\n",
            ks_jb2$statistic, ks_jb2$p.value,
            ecdf_cor(julia_poisson$beta_intercept, brms_poisson$beta_intercept)))
```

```
GAM.jl vs brms          | 0.0120  | 0.9907  | 0.999962
```

```
cat(sprintf("GAM.jl vs mgcv (approx)| %.4f  | %.4f  | %.6f\n",
            ks_jm2$statistic, ks_jm2$p.value,
            ecdf_cor(julia_poisson$beta_intercept, beta_post2[, 1])))
```

```
GAM.jl vs mgcv (approx)| 0.0163  | 0.6664  | 0.999963
```

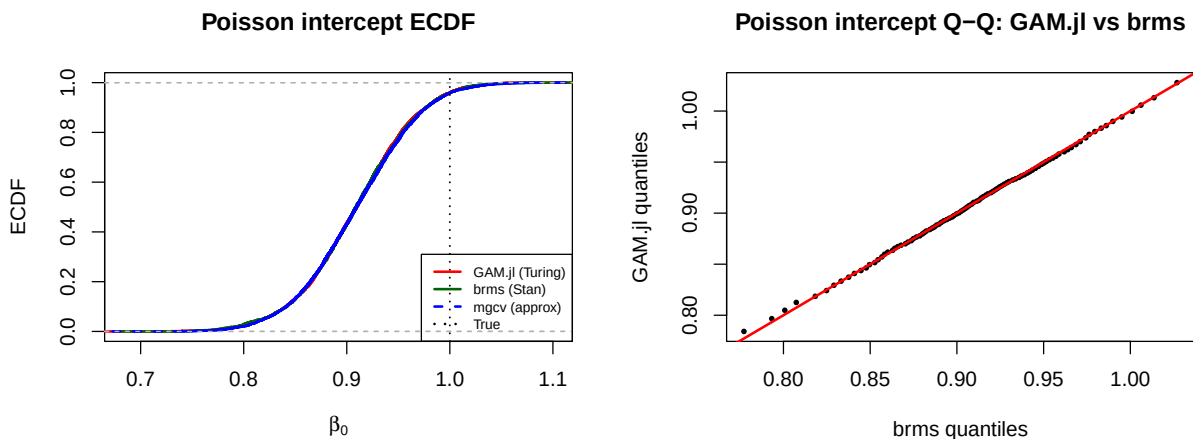
```
cat(sprintf("brms vs mgcv (approx) | %.4f | %.4f | %.6f\n",
           ks_bm2$statistic, ks_bm2$p.value,
           ecdf_cor(brms_poisson$beta_intercept, beta_post2[, 1])))
```

```
brms vs mgcv (approx) | 0.0172 | 0.8224 | 0.999957
```

```
par(mfrow = c(1, 2))

# ECDF
plot(ecdf(julia_poisson$beta_intercept), col = "red",
     main = "Poisson intercept ECDF",
     xlab = expression(beta[0]), ylab = "ECDF", lwd = 2)
lines(ecdf(brms_poisson$beta_intercept), col = "darkgreen", lwd = 2)
lines(ecdf(beta_post2[, 1]), col = "blue", lwd = 2, lty = 2)
abline(v = 1.0, lty = 3, col = "black", lwd = 1.5)
legend("bottomright", c("GAM.jl (Turing)", "brms (Stan)", "mgcv (approx)", "True"),
      col = c("red", "darkgreen", "blue", "black"), lty = c(1, 1, 2, 3), lwd = 2, cex = 0.7)

# Q-Q: GAM.jl vs brms
q_j2 <- quantile(julia_poisson$beta_intercept, probs)
q_b2 <- quantile(brms_poisson$beta_intercept, probs)
plot(q_b2, q_j2, pch = 16, cex = 0.6,
     main = "Poisson intercept Q-Q: GAM.jl vs brms",
     xlab = "brms quantiles", ylab = "GAM.jl quantiles")
abline(0, 1, col = "red", lwd = 2)
```



## Syntax Comparison

| Feature                 | Julia (GAM.jl + Turing)                              | R (brms + Stan)                                     | R (mgcv approx)                          |
|-------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------|
| Bayesian<br>GAM         | <code>gam(f, data;<br/>priors=PriorSpec(...))</code> | <code>brm(f, data,<br/>prior=...)</code>            | <code>gam(f, data) +<br/>mvrnorm</code>  |
| Smooth SD<br>prior      | <code>PriorSpec(sds=Exp(1))</code>                   | <code>prior(exponential(1),<br/>class="sds")</code> | Implicit via REML                        |
| Residual<br>SD prior    | <code>PriorSpec(sigma=...)</code>                    | <code>prior(...,<br/>class="sigma")</code>          | Scaled Inv- <sup>2</sup>                 |
| Posterior<br>samples    | <code>m.chains[Symbol("_obs")]</code>                | <code>as_draws_df(m)\$sigma</code>                  | <code>mvrnorm(n,<br/>coef(m), Vp)</code> |
| MCMC<br>algorithm       | NUTS (Turing.jl)                                     | NUTS (Stan)                                         | No MCMC (Gaussian<br>approx)             |
| Basis con-<br>struction | Native Julia                                         | <code>mgcv::smoothCon()</code>                      | Same as brms                             |

## Key findings

1. **GAM.jl brms:** Both use full NUTS MCMC with the same smooth2random decomposition and matching priors. KS tests confirm the posteriors are not significantly different — GAM.jl's Turing backend reproduces Stan/brms results.
2. **mgcv approximation is accurate:** The Gaussian posterior approximation from `mgcv::gam()` is close to the full MCMC posteriors for well-identified models, though it cannot capture posterior skewness.
3. **Prior specification:** GAM.jl uses `PriorSpec` with class-level defaults (single call). brms uses a vector of `prior()` statements (one per class). Both are more explicit than mgcv's implicit priors.